

Basic Reinsurance Terminology (review back of card)

- **Reinsurance:** Insurance on some/all of an insurer's risks
- **Ceding Company (CC):** Insurer that cedes risks to reinsurer
 - Aka direct insurer or cedant
- **Reinsurer:** Insurer that accepts (assumes) risk from CC
- **Treaty:** Legal agreement between CC and reinsurer
- **Retention:** Amount of risk CC retains (does not cede)
- **Retrocessionaire:** Reinsurer that accepts risk from another reinsurer
- **Cession:** Portion of risk transferred to reinsurer
- **Retrocession:** Portion of risk transferred from reinsurer to retrocessionaire

A Brief History of Reinsurance (review back of card)

- **Ancient origins (BCE)**

- ▶ Risk spreading on Chinese merchant ships, Babylonian maritime loans
- ▶ Greeks/Romans included risk fees on maritime loans

- **Medieval times**

- ▶ First insurance contract (1298), reinsurance contract (1370, marine)
- ▶ Term “reinsurance” first used in Florence (1457)

- **Modern developments**

- ▶ 18th-century England legislation strengthened insurable interest
- ▶ 19th-century fires spurred fire insurance, founding of reinsurers (e.g., Swiss Re)
- ▶ US life reinsurance began with American Central Life (1904)

- **Post-WWII to early 21st century**

- ▶ Reinsurance expanded post-WWII; N. American reinsurers ↓ since 1990s
- ▶ Remains vital in life/health, growing use in annuities

Compare recurring vs. portfolio reinsurance

- **Recurring:** Agreements on **new policies**
- **Portfolio:** Agreements to cede significant portions of **inforce policies** (aka block reinsurance)

Compare indemnity vs. assumption reinsurance

Indemnity (most common):

- Reinsurer reimburses CC for claims share
- No contractual relationship with p/h
- CC remains liable even if reinsurer defaults
- May include recapture provisions

Assumption:

- Reinsurer becomes directly liable to p/h
- CC no longer liable
- Requires novation (contract replacement)
- P/h must agree, receive assumption certificates

Compare proportional vs. non-proportional reinsurance

Proportional: Responsibility based on benefit known in advance

- Most common for life/health
- **Coinsurance:** Proportional sharing of risks/terms
- **Mod-Co:** Like coinsurance, CC retains reserves
 - Reinsurer pays reserve $\uparrow$ net of inv income
- **YRT:** Transfers only mortality/morbidity risk in NAR

Non-proportional: Liability depends on claims incurred

- More common for unpredictable claims (P&C)
- **Stop Loss:** Covers losses $>$ attachment point
- **Catastrophe:** High-severity, low-frequency events

Compare automatic vs. facultative reinsurance

- **Automatic:** Risks exceeding retention that meet UW criteria are automatically ceded
 - CC **must cede**, reinsurer **must accept**
- **Facultative:** CC **may choose to cede** or not
 - Reinsurer **must approve each risk**
- **Variations:**
 - Conditional automatic: Reinsurer provides UW service
 - Facultative obligatory: Reinsurer can only decline if it hits full retention on that individual

Compare excess treaty vs. quota share reinsurance

- **Excess treaty:** Covers risks based on scheduled retention limits
- **Quota share: Fixed %** of each risk is ceded
 - Typically applies to **first dollar** of coverage
 - Can be combined with excess risk management

Compare experience rated vs. non-experience rated reinsurance

- **Experience Rated:** CC shares in reinsurer profits via **experience refund**
 - Higher reinsurance premiums
- **Non-Experience Rated:** More common today

Compare traditional vs. financial reinsurance

- **Traditional:** Primary purpose is **risk transfer**
- **Financial:** Primary purpose is **specific business objective** (e.g., surplus ↑, tax ↓, acquiring blocks), risk transfer secondary
 - Distinctions may require close examination of risks transferred

List uses of reinsurance

Risk management:

- Mortality/morbidity transfer
- Lapse/surrender transfer
- Investment risk transfer

Business/financial strategy:

- New business financing, UW assistance
- Entering new markets, M&A, divesting product lines
- Optimizing admin functions
- Reinsurer returns/inforce ↑
- Surplus management/optimization
- Redundant reserve management

How can reinsurance transfer risk?

Mortality/morbidity (oldest, most common):

- Life: Excess/QS treaties; retention limit key to limiting insolvency
- Deferred annuities: Usually % of fund reinsured
- Payout annuities: Rare (flat retention or QS)

Lapse/surrender (relevant when):

- Significant first-year strain
- Premiums ↑ sharply
- Level premiums, claims ↑
- Anti-selection/disintermediation from high lapses

Investment:

- Leverage reinsurer capabilities, limit concentrations
- Common with coinsurance/mod-co

How does reinsurance aid new business/market entry?

- **New business financing:** Relieves surplus strain
- **UW assistance:** Manuals, training, audit, difficult case review
- **Market entry:** Expertise on products, regulation, business lines
- **Fronting:** Direct insurer issues policies, reinsures substantially all risk
 - ▶ Reinsurer handles admin
 - ▶ Common in P&C/A&H; rare for ILA/health

Why might CC cede large portions of new business?

- Low marginal cost (reinsurer's lower expenses or different assumptions)
 - Small profit > no profit
- Allows writing larger policies
- UW gains experience from reinsurer
- Recapture provisions → future profit ↑

How does reinsurance relate to divestiture, M&A, and admin optimization?

- **Divesting product line:** Exit lines/areas, reallocate capital, simplify admin, overhead ↓
- **M&A:** Transfer blocks or sell/purchase companies
 - Assumption typically used for permanent sales; indemnity sometimes preferred
- **Admin optimization:**
 - Cede business → admin burden ↓
 - Assume business → use excess admin capacity

How does reinsurance benefit reinsurers?

- **Returns** ↑: Assume business for profit and ROI
- **Inforce** ↑: Leverage under-utilized admin capacity, spread overhead, augment direct sales

What is the purpose of producer-owned reinsurance companies (PORCs)?

Source: Tiller Ch. 1: Basic Methods of Reinsurance

- Substantial reinsurance volume ceded to PORCs
- Allows high-producing agent groups to participate in earnings
- Agent loyalty ↑, rewards high production

How does reinsurance optimize surplus and fund redundant reserves?

Surplus optimization:

- Fund acquisition costs from high sales
- Meet RBC/statutory/rating requirements
- Replenish surplus post-poor experience
- Meet dividend/debt service objectives

SPVs (form of captive):

- Raise capital, mitigate risks via securitization
- Securitization = selling cash flow streams (premiums, claims)
- Major use: Fund redundant XXX/AXXX reserves

Describe YRT reinsurance and which products it suits.

Reinsures mortality/morbidity risk only

CC pays:

- Reins premium to reinsurer
- All p/h benefits and expenses

Reinsurer pays:

- Reinsured claims (reimburses CC)
- Expense allowance

CC takes small ceded reserve credit (= unearned YRT premium)

Good for:

- Term, WL, UL (mortality)
- Disability, LTCI, critical illness (morbidity)

Not good for:

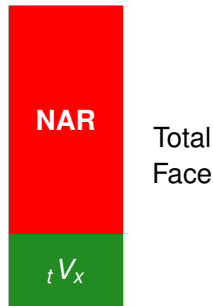
- Lapse/investment risk transfer
- Annuities

Provides little surplus relief vs. coinsurance

How is ceded NAR calculated under YRT?

NAR = Face – Terminal Reserve (or per treaty definition)

- Calculated at policy level
- Ignore deficiency reserves
- YRT Premiums = YRT Rates $\times$ Ceded NAR
- Ceded DB = Reinsurer's NAR share
- NAR must be well documented in treaty
 - ▶ Companies agree on method
 - ▶ DB in calculation = DB paid on death
 - ▶ NAR used $\approx$ actual NAR



List 4 YRT retention methods.

1. **Pro Rata** (preferred)

- ▶ Constant reinsured % = $\frac{\text{Orig Face Ceded}}{\text{Total Orig Face}}$
- ▶ → Very small reins amounts in later durations

2. **Level/Constant Retention**

- ▶ CC retains fixed NAR amount
- ▶ Reinsured NAR ↓ over time

3. **Constant Risk Reinsured** (rare)

- ▶ Reinsured amount = constant
- ▶ Retained NAR ↓ over time

4. **Formula Retention**

- ▶ NAR and retention per agreed formula

How are YRT reinsurance premiums calculated?

$$\text{ReinsPrem} = \text{YRT Rate} \times \frac{\text{Ceded NAR}}{1000} + \text{Cession Fee}$$

- YRT premium rate = multiple of mortality table
 - Vary by age, duration, sex, UW class
 - Select periods common (e.g., 15 years)
- May have zero first-year premium (ZFT)
- Calendar year scales avoid year-end reins reserves
- Substandard ratings:
 - Table rating: ↑ by multiple (e.g., 150%)
 - Flat extra: May have allowance on extra premium
- Riders: Accidental death may be reinsured separately, term riders use base
- Monthly renewable term (MRT) for UL: YRT done monthly

Describe coinsurance and compare to YRT.

Reinsurer has proportionate share of:

- Direct p/h premium
- Direct reserve
- All benefits (mortality, morbidity, lapse, surrender, investment)

vs. YRT:

- Cedes more premium, reserves, benefits
 - Reinsurer must track all these
 - EAs more complicated
- Can be used for any insurance/annuities
- More surplus relief than YRT
- Passes deficiency reserves to reinsurer

How are coinsurance premiums calculated? List approaches for banded premiums.

$$\text{ReinsPrem} = (\text{Amount Ceded Per 1000}) \times (\text{GP Per 1000})$$

Approaches for banded premiums:

1. Simplest
 - ▶ Use actual GP per policy
 - ▶ Common allowances for all bands
2. More equitable (more complicated)
 - ▶ Use actual GP per policy
 - ▶ Vary allowances by band
3. Single band
 - ▶ Use GP for specific band regardless of actual
 - ▶ One set of allowances for all policies

Describe variations in coinsurance expense allowances.

- Allowance = % of ceded premium
- Uniquely developed per product
- Varies by duration, sex, risk class, band, etc.
 - Based on reinsurer profit objectives
- Premium tax methods:
 - Most common: Reinsurer includes provision in basic allowance
 - Reimburse exact amount (challenging)
 - Pay fixed amount as proxy

How are the following treated under coinsurance:

- Policy fee
- Policyholder dividends
- Policy loans

- **Policy fee:** May be fully retained by CC
 - Reinsurer may coinsure so it's included in deficiency reserve
- **P/h dividends:** Reinsurer may or may not participate
 - If no participation → asset mgmt challenges for CC
- **Policy loans:** No reinsurer participation ever
- Substandards and riders treated similarly to YRT

Compare mod-co to regular coinsurance.

Mod-co = coinsurance except:

1. CC retains assets backing ceded reserves
2. No ceded reserve credit on CC balance sheet
3. No reinsured reserve on reinsurer balance sheet
4. Reinsurer pays mod-co adjustment to CC
 - ▶ Main complexity vs. coinsurance

Advantages vs. coinsurance:

- CC has more investment control
- Good for interest-sensitive liabilities, p/h dividends
- More assets for policy loans
- Good when reinsurer offshore/unauthorized
 - ▶ No reserve credit questions

How is the mod-co adjustment calculated?

Positive mod-co adjustment = payment by reinsurer

Mod-co adjustment =

Ending ceded stat reserves

- Beginning ceded stat reserves
- “Interest” on beginning ceded stat reserves

Mod-co interest rate must be clearly defined in treaty

- Should ensure reinsurer participates in total asset performance
 - Asset defaults, capital gains/losses, timing risks, inv income
- Not simple interest rate in practice
- Governed by NAIC Life & Health Reins Agreements Model Reg

List goals and advantages of advanced reinsurance.

Goals:

1. Surplus relief to CC
2. Reduce asset transfers between companies

Advantages for CC:

1. Allow ceded reserve credit for non-admitted reinsurers
2. Improve security of ceded reserve credit
3. Coordinate asset mgmt for asset-sensitive liabilities
4. Minimize capital gains/losses at inception
5. Easier recapture (minimizes capital gains/losses)

How is the outstanding surplus account calculated?

OSA tracks reinsurer's investment in CC

$$OSA_t = OSA_{t-1} + \text{StatGain}_t - i \times OSA_{t-1} - \text{RiskCharge}_t$$

- StatGain = reinsurer's gain from operations (profit)
 - Initial OSA = Initial Stat Gain < 0
- $i \times OSA_{t-1}$ = inv income on beginning OSA
- RiskCharge = % of beginning OSA

OSA calculation same for all methods

Compare advanced reinsurance forms regarding:

- Reinsurer holds reserve?
- CC takes reserve credit?
- Reinsurer holds assets backing ceded reserves?
- Reinsurer owns performance of assets?
- Calculate mod-co adjustment?
- Establish FW balance instead of exchanging cash?
- Establish FW balance for initial allowance only?
- Complies with state laws?

Source: Tiller Ch. 5: Advanced Methods of Reinsurance

	1. Coins.	2. Mod-Co	3. FW-Co	4. FW-Mod-Co
Reinsurer holds reserve	Yes	No	Yes	No
Ceding company takes reserve credit	Yes	No	Yes	No
Reinsurer holds assets backing ceded reserves	Yes	No	No	No
Reinsurer owns performance of assets	Yes	No	Yes	Yes
Calculate mod-co adjustment	No	Yes	No	Yes
Establish a FW balance instead of exchanging cash	No	No	Yes	Yes
Establish a FW balance for initial allowance only	No	No	No	Yes
Complies with state laws	Yes	Yes	Yes	?

5. Part-co is a blend of regular coinsurance and mod-co

Compare reinsurance report vs. FW balance calculation.

Net Cash Due to Reinsurer =

- Ceded premium
- Allowances
- Ceded benefits
- Mod-co adjustment (if any)
- Experience refund (if any)

EOY FW Balance = BOY FW Balance

- + Ceded premium
- Allowances
- Ceded benefits
- Mod-co adjustment (if any)
- Experience refund (if any)
- Risk charge

FW arrangement: No cash transferred between companies (except risk charge may be paid in cash)

Compare FW coinsurance vs. regular coinsurance.

Differences:

1. No cash exchanged at inception/renewal
2. Net cash owed → accounts payable/receivable
 - ▶ Net cash owed > 0 → accounts payable liability
 - ▶ Otherwise → accounts receivable asset
 - ▶ One company's receivable = other's payable
3. Assets stay with CC (like mod-co)
 - ▶ But CC shows ceded reserve credit (like regular coinsurance)
 - ▶ FW balance adjusted for interest on retained assets
 - ▶ Authors include ceded interest in ceded premium
4. Alternative OSA calculation for FW:

$$\begin{aligned} \text{OSA}_t &= \text{OSA}_{t-1} + \text{CededPrem}_t \\ &\quad - \text{CededBen}_t - \text{CededResIncr}_t - \text{ExperienceRefund}_t - \text{RiskCharge}_t \end{aligned}$$

Describe FW mod-co.

Tip: Think of it as “allowance retained mod-co”

FW-Mod-Co = mod-co except:

- Reinsurer retains initial allowance
- Reinsurer's initial FW liability = initial allowance
- CC's initial FW asset = initial allowance

FW-Mod-Co may violate state laws

- State laws usually require quarterly settlement
- May be possible to get approval

Describe part-co.

Part-co = blend of regular coinsurance and mod-co

- **Main advantages:**

- ▶ Eliminates/minimizes cash transfer
- ▶ No FW balance to track

- **Cash owed by each party cancels at inception:**

- ▶ Initial coinsurance reserve = initial allowance
- ▶ Remaining reserves are mod-co reinsured

- **Co/mod-co split adjusted in renewals to minimize cash flow:**

- ▶ Generally, stat profits ↓ coinsurance portion
- ▶ Could be designed so coinsurance portion = reinsurer's outstanding investment

- **Disadvantages:**

- ▶ Very complicated to administer/understand
- ▶ Regulators may be concerned about ↑ coinsurance portion
- ▶ Must calculate stat gain twice: preliminary then final

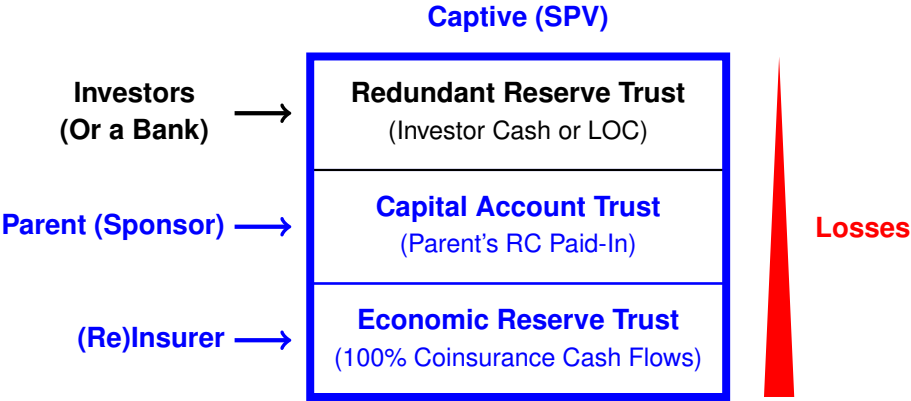
What are SPV objectives?

- **SPV owner = “sponsor”** (reinsurer, direct insurer, or parent)
- **Sponsor objectives: Fund redundant reserves with low-cost capital**
 - ▶ Allows other capital for higher risk (higher returns)
 - ▶ Redundant Reserves = Total Stat Reserves – Economic Reserves
 - ▶ Total Stat Reserves = Basic Stat Reserve + XXX Reserves
 - ▶ Economic Reserves = GAAP BenRes – DAC
 - ▶ Should emerge from cash flows without additional funding
- **Investor objectives: Low-risk investment**
 - ▶ Low loss possibility
 - ▶ Willing to accept low return (low risk)
 - ▶ Investors: Bondholders or bank issuing LOC

Describe SPV capital structure.

- **“Required Capital” = capital contributed by sponsor**
 - Greater of min regulatory capital and capital required by investors
 - May not tie to regulatory/rating agency capital
- **Economic Reserves + Required Capital**
 - Should fund claims/expenses even under adverse developments
 - Higher RC = less risk for investors
 - If RC too high → SPV may not be viable

List the 3 trust types in typical securitization SPV model.



List steps for structuring an SPV.

1. Sponsor forms captive, contributes RC

2. If issuing bonds, document:

- ▶ Restrictions on use of funds
- ▶ Delineation between reserve/capital requirements
- ▶ Investor funds used only after reserves/RC
- ▶ How/when dividends, cash, etc. may be paid to sponsor
- ▶ Any experience refunds
- ▶ Interest payment on bonds
- ▶ Repayment of initial principal

3. Place funds in 3 trusts:

- ▶ Redundant reserve trust
- ▶ Capital account trust
- ▶ Economic reserve trust

List SPV investor considerations.

- SPVs expensive to establish: Requires sufficient volume
- Bonds no longer issued post-crisis
- Structures usually include NB for limited time (e.g., 1 year)
- SPVs usually cover limited product groups (e.g., level term)
- Max funding required for RC/redundant reserves at inception
 - But could have agreement for more in future
- Usually based on some form of coinsurance
- CC must be able to take ceded reserve credit (hence trusts)
- Assets in SPV for long time (30+ years)
- Investor retains inv income from assets

List advantages of using LOCs for SPVs.

- **Only annual redundancy amount needs LOC funding**
- **Bank ↑ LOC amount each year for guaranteed fixed charge**
 - Avoids refinancing the LOC
 - Better than bonds (would need reissue/refinance)
- **Long-term LOC costs ↓ as banks gain comfort**